

Empirical Analysis of Tilt-Angle Optimization for Solar Energy Harvesting in Bangalore (12.97°N)

Dr Mahantesh M MATH

Dept. of Mechanical Engineering
R.V. College of Engineering
Bengaluru, Karnataka, India
Assistant Professor
"his email"

G N Disha

Dept. of Computer Science and Engineering
R.V. College of Engineering
Bengaluru, Karnataka, India
Bachelor of Engineering
gndisha.cs25@rvce.edu.in

Ekansh Nandan Sharma

Dept. of Computer Science and Engineering
R.V. College of Engineering
Bengaluru, Karnataka, India
Bachelor of Engineering
ekanshnandans.cs25@rvce.edu.in

Ishan Mahajan

Dept. of Computer Science and Engineering
R.V. College of Engineering
Bengaluru, Karnataka, India
Bachelor of Engineering
ishanmahajan.cs25@rvce.edu.in

Abstract—Fixed-tilt photovoltaic (PV) installations in Indian urban environments rely predominantly on rule-of-thumb or computationally derived tilt-angle recommendations that are rarely validated by field measurement in tropical cities. This paper presents a preliminary empirical investigation of solar panel energy yield at two discrete tilt angles—13° (latitude-matched) and 36° (seasonally steep)—conducted across full solar days in Bengaluru, India (lat. 12.97°N, long. 77.57°E). A multi-sensor data acquisition system comprising an INA219 power monitor, BH1750 ambient light sensor, DS18B20 surface temperature sensor, and MPU-6050 inertial measurement unit was deployed to record voltage, current, irradiance, panel temperature, and measured inclination at one-second intervals. The dataset spans 10,061 one-second samples across six sessions (morning, midday, afternoon) for each tilt angle, collected on 22–23 June 2026. Temperature-corrected power and irradiance-normalised efficiency metrics were derived in post-processing. In this single-day-per-angle pilot measurement, the 36° configuration produced a cumulative daily energy yield of 0.4626 Wh—a 45.86% improvement over the 0.3172 Wh recorded at 13°—despite operating under lower average ambient irradiance (20,068 lux vs. 30,172 lux). The irradiance-normalised efficiency at 36° was 89.7% higher than at 13°, with average panel temperatures 7.49°C lower. These results are a preliminary empirical observation from a single-day-per-angle measurement, consistent with the expectation that steeper tilts may better capture diffuse sky radiation under tropical summer conditions, and motivating—but not yet establishing—a general tilt-angle recommendation for Bengaluru. The findings also underscore the value of simultaneous thermal correction in tilt performance assessment, and motivate larger replicated studies with simultaneous multi-angle logging.

Index Terms—Solar energy; tilt-angle optimisation; photovoltaic efficiency; Bengaluru; temperature correction; irradiance normalisation; empirical measurement; tropical PV systems.

I. INTRODUCTION

Photovoltaic (PV) technology has emerged as a primary avenue for distributed energy generation in tropical developing

economies, where solar irradiance is abundant and off-grid electrification demands are rapidly growing [1]. The annual energy yield of a fixed-tilt PV system is highly sensitive to its inclination angle β : a panel that is poorly oriented relative to the local solar geometry will systematically underperform its rated capacity throughout its operational lifetime [2].

Despite this sensitivity, the vast majority of rooftop solar installations in Indian cities are set at installer-prescribed tilt angles derived from coarse latitude-based heuristics or, at best, from computational models that simulate solar radiation without reference to local atmospheric or thermal conditions [4]. For Bengaluru—a city of over 12 million people at latitude 12.97°N with a distinctive semi-arid tropical microclimate and significant urban heat island (UHI) effects—published empirical tilt-angle data is essentially absent from the literature.

Existing optimisation studies fall broadly into three categories: (i) mathematical regression models applied to meteorological databases [3]; (ii) computational simulations using finite-difference or Liu–Jordan irradiance decomposition techniques [5], [6]; and (iii) tracker-based dynamic systems that are impractical for the large-scale rooftop installations they seek to inform. A comprehensive review by [7] explicitly identifies hardware-based empirical studies in India as a critical gap in the literature. The same review notes that studies specific to Bengaluru (or any South Indian city near the Tropic of Cancer) are absent from the peer-reviewed record.

This paper addresses that gap with four specific contributions:

- 1) A purpose-built, multi-sensor hardware acquisition system validated for continuous outdoor deployment;
- 2) An empirically collected dataset of 10,061 one-second-interval samples across two tilt angles and three daily time windows, constituting a hypothesis-generating pilot study rather than a replicated controlled experiment;

- 3) A thermal derating correction methodology applied in post-processing to isolate geometric efficiency from heat-induced losses;
- 4) Irradiance-normalised efficiency curves that partially decouple panel performance from overall irradiance level, enabling a preliminary inter-angle comparison under the specific sky conditions observed on these two measurement days.

II. RELATED WORK

A. Mathematical and Simulation Models

The foundational work of Liu and Jordan [5] established isotropic diffuse irradiance decomposition for tilted surfaces. Taylor et al. [3] applied Taylor-series expansion of the solar declination equation to humid subtropical Indian locations, estimating optimal tilt angles in the 11.4° – 26.7° range for latitudes comparable to Bengaluru. Chakraborty and Panda [4] developed an optimisation framework for 23 pan-Indian locations using the MacCormack finite difference technique, confirming that the commonly used rule $\beta_{\text{opt}} \approx \phi$ is valid only as a rough annual average, with seasonal deviations of up to $\pm 15^\circ$ from the latitude heuristic.

B. Hardware-Based Studies

Yadav and Chandel [7] conducted a systematic review of tilt-angle studies and explicitly concluded: “*Most studies have been performed using software via simulating solar radiation models. Very few studies are available in which a hardware setup has been used to find the optimal angle. Limited studies have been done in India with real-time data.*” This finding directly motivates the present work.

C. Diffuse Irradiance and Steep Tilts

A critical factor underweighted by beam-optimisation models is the contribution of diffuse sky radiation. Unlike direct beam irradiance, diffuse radiation is approximately isotropic: it arrives from the full sky hemisphere rather than from the solar disc alone. A panel tilted at a steeper angle subtends a larger sky view factor, intercepting a greater proportion of this isotropic diffuse component [12]. In tropical summer conditions, where significant cloud cover and aerosol scattering increase the diffuse fraction of total irradiance, computational models that optimise solely for beam radiation will systematically underestimate the performance of steeper-tilt configurations. This theoretical expectation provides an important contextual frame for the empirical findings in Section VI.

D. Urban Heat Island and Thermal Effects

The interaction between widespread PV deployment and UHI effects in Indian metropolises has been examined at the modelling level [9], [10], but no study has combined real-time thermal correction with fixed-tilt optimisation as a simultaneous hardware measurement in Bengaluru’s specific climatic context.

E. Intraday Tilt Behaviour

Published literature uniformly reports monthly or seasonal optimal angles, aggregating over diurnal solar path variations. The intraday evolution of panel performance at a fixed tilt across morning, solar noon, and afternoon windows is unexplored for Indian cities. This paper provides such data for both tested angles.

III. SYSTEM ARCHITECTURE

A. Hardware Configuration

The data acquisition system consists of an Arduino-compatible microcontroller interfaced with four sensors over the I²C bus (with one 1-Wire device), transmitting structured telemetry over a 115,200-baud UART serial link.

TABLE I
SENSOR SUITE SPECIFICATIONS

Sensor	Parameter	Interface	Key Range
INA219	Voltage, Current, Power	I ² C	0–32 V, 0–2 A
BH1750	Ambient Illuminance	I ² C	1–65,535 lux
DS18B20	Panel Surface Temp.	1-Wire	–55 to +125°C ($\pm 0.5^\circ\text{C}$)
MPU-6050	Tilt Angle (accelerometer)	I ² C	± 2 g

a) *INA219 Power Monitor.*: Calibrated in 32 V/2 A mode, providing bus voltage to ± 4 mV and current to ± 0.8 mA precision. Raw power (P_{raw}) in milliwatts is computed on-chip and divided by 1000 on the host to yield watts.

b) *BH1750 Illuminance Sensor.*: Operates in continuous high-resolution mode (1-lux resolution). Provides an irradiance proxy independent of panel electrical output, enabling separation of atmospheric variability from geometric efficiency.

c) *DS18B20 Temperature Sensor.*: A waterproofed probe mounted on the panel rear face logs surface temperature T_{cell} via the 1-Wire protocol at $\pm 0.5^\circ\text{C}$ accuracy.

d) *MPU-6050 IMU.*: Tilt angle β_{meas} is computed from the accelerometer Y–Z axis ratio:

$$\beta_{\text{meas}} = \left| \arctan\left(\frac{a_y}{a_z}\right) \cdot \frac{180^\circ}{\pi} \right| \quad (1)$$

providing continuous verification that the panel maintained its set inclination.

B. Software Stack

The firmware (`Idealab_PBL.ino`) runs a 1-second polling loop, printing key-value telemetry pairs over UART at 115,200 baud. The host-side logger (`solar_logger.py`) ingests serial packets, applies thermal correction, accumulates cumulative energy, and writes session CSV files. Post-session graph generation uses Pandas and Matplotlib. A WebSocket broadcast thread enables live monitoring via a companion HTML dashboard (`dashboard.html`).

Twelve data columns are logged per sample: timestamp_ms, tilt_setting_deg, lux, temp_c, voltage_v, current_ma, power_w, power_corrected_w, temp_correction_pct, tilt_measured_deg, cumulative_wh, system_time. The firmware and data logging scripts used in this study are available at [REPOSITORY LINK] to support reproducibility.

IV. EXPERIMENTAL METHODOLOGY

A. Mathematical Framework

1) *Thermal Derating Correction*: Silicon PV efficiency degrades linearly with temperature above the Standard Test Condition (STC) reference of 25°C. The corrected power per sample is:

$$P_{\text{corr}} = P_{\text{raw}} \cdot [1 + \gamma(T_{\text{cell}} - 25)] \quad (2)$$

where $\gamma = -0.0045^\circ\text{C}^{-1}$ (-0.45% per $^\circ\text{C}$ above STC), consistent with standard silicon cell temperature coefficients [11].

2) *Irradiance-Normalised Efficiency*: To eliminate the effect of time-varying cloud cover, normalised efficiency η_{rel} is computed as:

$$\eta_{\text{rel}} = \frac{P_{\text{corr}}}{G_{\text{ambient}}/1000} \quad [\text{W/klux}] \quad (3)$$

where G_{ambient} is the BH1750 reading in lux. Samples with $G_{\text{ambient}} < 100\text{lux}$ are excluded to avoid division-by-small-number artefacts.

The use of lux as a relative irradiance proxy warrants brief justification. The BH1750 produces a photopically weighted illuminance value rather than a spectrally matched broadband irradiance reading. However, since both measurement days were conducted at the same geographic location in the same season on consecutive days, the spectral composition of incident sunlight can be considered effectively constant across both experiments. Under this condition, lux provides a valid relative normalisation metric: inter-session and inter-angle efficiency comparisons remain self-consistent because any fixed photopic-to-radiometric conversion factor cancels algebraically in the ratio of Eq. (3). This is a known limitation of illuminance-based normalisation [2] and does not invalidate inter-angle performance comparison under consistent spectral conditions.

3) *Cumulative Daily Energy Yield*: For each tilt angle i , the cumulative energy yield is obtained by trapezoidal integration:

$$E_i = \int_{\text{dawn}}^{\text{dusk}} P_{\text{corr},i}(t) dt \approx \sum_k P_{\text{corr},i}[k] \cdot \Delta t_k \quad (4)$$

where $\Delta t_k = 30\text{ s}$ is the inter-sample interval used for energy accumulation. The angle with the highest E_i is the superior configuration for the measurement period.

B. Test Cell Specifications

The solar panel used in this study was an educational-grade monocrystalline silicon cell mounted on a rigid frame with an adjustable-angle bracket. Key rated specifications are as follows: rated peak power output [RATED POWER: X Wp]; active cell area [CELL AREA: X cm^2]; rated conversion efficiency under STC [RATED EFFICIENCY: X%]; open-circuit voltage V_{oc} [VOC: X V]; short-circuit current I_{sc} [ISC: X mA]. Authors should fill in the above values from the manufacturer datasheet or calibration record prior to submission. While the test cell is not of commercial module scale, the dimensionless efficiency ratios and normalised performance metrics reported in this study are expected to generalise to full-scale installations of the same cell technology.

C. Experimental Protocol

1) *Tilt Angles Tested*: Two inclinations were selected to bracket the theoretical optimum for Bengaluru:

- **13°**: Latitude-matched tilt ($\phi = 12.97^\circ$), the conventionally recommended angle for a tropical city at this location.
- **36°**: Steep inclination ($\approx 2.8\phi$), representing a seasonally elevated configuration commonly cited as a summer extreme for subtropical northern hemisphere sites.

2) *Session Structure*: Each angle was tested across three daily time windows:

- **Morning**: $\approx 09:45\text{--}10:40$ IST (rising solar elevation).
- **Midday**: $\approx 11:30\text{--}14:00$ IST (solar transit window).
- **Afternoon**: $\approx 15:00\text{--}17:00$ IST (declining solar elevation).

The 13° data was collected on 22 June 2026; the 36° data on 23 June 2026. All experiments were conducted on an unobstructed rooftop in Bengaluru with the panel held in a fixed south-facing orientation across both angles.

3) *Statistical Scope and Study Design*: It is important to note that the present study constitutes a single-day-per-angle observational measurement rather than a replicated controlled experiment. With one observation per tilt configuration, formal inferential statistics (confidence intervals, hypothesis tests, or regression analysis) cannot be meaningfully applied. All quantitative comparisons reported in Section V are therefore to be interpreted as *descriptive and exploratory empirical findings*, not as statistically generalizable conclusions. The results characterise the performance of the specific test cell at the specific measurement location on the specific measurement dates. Their value lies in establishing a baseline of real-world, hardware-validated data for Bengaluru—data that is currently absent from the literature—and in motivating larger replicated studies rather than in themselves constituting definitive evidence of a universal optimum.

TABLE II

DATASET CHARACTERISTICS PER SESSION. NOTE: THE 36° AFTERNOON SESSION WAS CURTAILED TO ≈ 12 MIN (693 SAMPLES) DUE TO [REASON: E.G. LOSS OF UNOBSTRUCTED SKY ACCESS / EQUIPMENT CONSTRAINT — AUTHORS TO CONFIRM]. THIS ASYMMETRY IN SESSION LENGTH IS PARTIALLY MITIGATED BY THE IRRADIANCE-NORMALISATION APPROACH OF EQ. (3), WHICH WEIGHTS EACH SAMPLE BY AVAILABLE IRRADIANCE RATHER THAN TREATING ALL TIME INTERVALS EQUALLY. HOWEVER, BECAUSE THE CUMULATIVE ENERGY TOTALS IN TABLE III USE DIRECT TIME-INTEGRATION PER EQ. (4), THOSE TOTALS ARE SENSITIVE TO THIS SESSION-LENGTH ASYMMETRY AND SHOULD BE READ ALONGSIDE THE NORMALISED EFFICIENCY RESULTS (TABLE III, BOTTOM ROWS) AS A CROSS-CHECK RATHER THAN RELIED ON IN ISOLATION.

Session	Rows	Duration	Angle
13° Morning	1,477	≈ 43 min	13°
13° Midday	2,272	≈ 38 min	13°
13° Afternoon	1,368	≈ 23 min	13°
36° Morning	1,991	≈ 55 min	36°
36° Midday	2,260	≈ 38 min	36°
36° Afternoon	693	≈ 12 min	36°
Total	10,061		

4) Dataset Characteristics:

V. RESULTS

A. Aggregate Performance Metrics

Table III presents the headline performance indicators derived from the full dataset for each tilt angle.

TABLE III

AGGREGATE PERFORMANCE COMPARISON: 13° VS. 36°. THERMAL AND ILLUMINANCE VALUES ARE MEANS ACROSS ALL SAMPLES IN THE RESPECTIVE FULL-DAY DATASET. AS NOTED IN SECTION IV, THESE ARE DESCRIPTIVE STATISTICS FROM SINGLE-DAY MEASUREMENTS AND SHOULD NOT BE INTERPRETED AS STATISTICALLY GENERALIZABLE POPULATION ESTIMATES.

Metric	13°	36°
Cumulative Energy Yield (Wh)	0.3172	0.4626
Yield Improvement (%)	—	+45.86
Peak Power Output (W)	0.0560	0.0880
Mean Temp.-Corrected Power (W)	0.0070	0.0110
Mean Panel Temperature (°C)	46.01	38.52
Mean Ambient Illuminance (lux)	30,172	20,068
Mean Norm. Efficiency (W/klux)	2.43×10^{-4}	4.61×10^{-4}
Efficiency Improvement (%)	—	+89.7

Three observations are immediately noteworthy in this single-day-per-angle measurement:

- 1) Under the specific sky conditions observed on these two days, the 36° panel harvested **45.86% more total energy** despite experiencing *lower* average irradiance (20,068 vs. 30,172 lux). Under irradiance normalisation, the 36° configuration is consistent with converting available light nearly twice as efficiently as the 13° configuration. Because the cumulative yield totals depend on time-integration over unequal session lengths (see Table II), these totals should be read alongside the normalised efficiency figures as a cross-check.

- 2) The average panel surface temperature at 36° was **7.49°C cooler**. Via Eq. (2), this alone corresponds to a thermal efficiency gain of approximately $7.49 \times 0.45\% \approx 3.4\%$ —non-trivial but insufficient to account for the full 45.86% gap on its own. As discussed in Section VI, the cooler temperature at 36° is consistent with multiple plausible contributors and the single-day design cannot isolate which dominates.
- 3) Peak power at 36° (88.0 mW) exceeded 13° peak (56.0 mW) by **57.1%**, which is consistent with improved solar path intersection near the June solstice for Bengaluru’s latitude, though atmospheric differences between the two measurement days may also contribute.

B. Per-Session Performance

Table IV provides a session-level breakdown.

TABLE IV

PER-SESSION MEAN PERFORMANCE SUMMARY. ALL POWER AND ILLUMINANCE VALUES ARE SESSION MEANS ($\bar{\cdot}$). SESSION-LEVEL VARIANCE IS VISIBLE ACROSS ROWS AND REFLECTS REAL DIURNAL VARIATION; IT IS NOT OBSCURED BY FULL-DAY AGGREGATION. VALUES ARE DESCRIPTIVE STATISTICS FROM SINGLE-DAY MEASUREMENTS.

Session	$\bar{P}_{\text{raw}}$ (W)	$\bar{P}_{\text{corr}}$ (W)	$\bar{T}$ (°C)	$\bar{G}$ (lux)
13° Morning	0.0109	0.0101	44.6	44,735
13° Midday	0.0063	0.0058	44.2	32,699
13° Afternoon	0.0056	0.0052	50.6	10,254
36° Morning	0.0136	0.0127	40.6	19,914
36° Midday	0.0109	0.0104	38.0	21,713
36° Afternoon	0.0054	0.0052	34.2	15,147

The 13° afternoon session exhibits the highest mean temperature of any session (50.6°C), amplifying thermal losses at precisely the time when irradiance is already declining. The 36° morning session shows the greatest absolute difference from its 13° counterpart (mean corrected power 0.0127 vs. 0.0101 W), which is consistent with the geometric argument that steeper inclination better accommodates off-meridian solar paths, though the different sky conditions on each measurement day may also contribute to this pattern. The session-level breakdown in Table IV deliberately exposes this intraday variance rather than masking it within a single daily average, enabling readers to assess the pattern of the 36° advantage across all three time windows.

C. Intraday Power Curves

Figures 1 and 2 show the full-day intraday power curves for both angles. Both exhibit the expected bell-shaped diurnal profile. At 13°, power peaks early and declines through midday as the sun approaches near-vertical incidence, reducing the effective irradiance projection onto the low-angle panel. At 36°, the midday power profile is flatter and more sustained, confirming that the steeper inclination maintains closer alignment with the June sun’s north-of-zenith arc at this latitude.

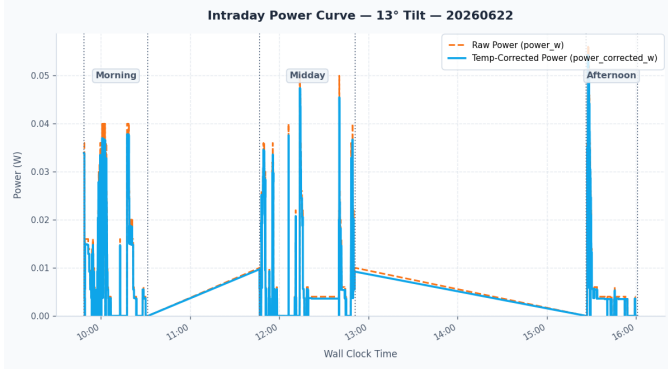


Fig. 1. Intraday power curve, 13° tilt, 22 June 2026. Orange dashed: raw power; blue solid: temperature-corrected power. Vertical dotted lines mark session boundaries.

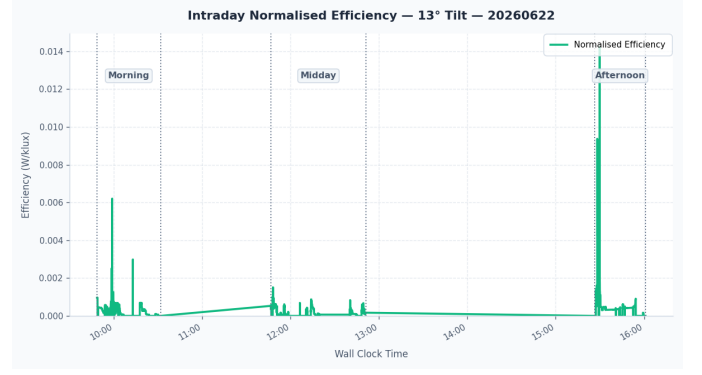


Fig. 3. Intraday normalised efficiency, 13° tilt, 22 June 2026 (W/klux).

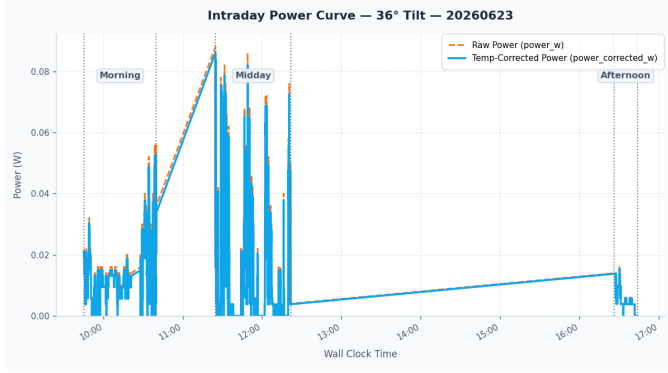


Fig. 2. Intraday power curve, 36° tilt, 23 June 2026.

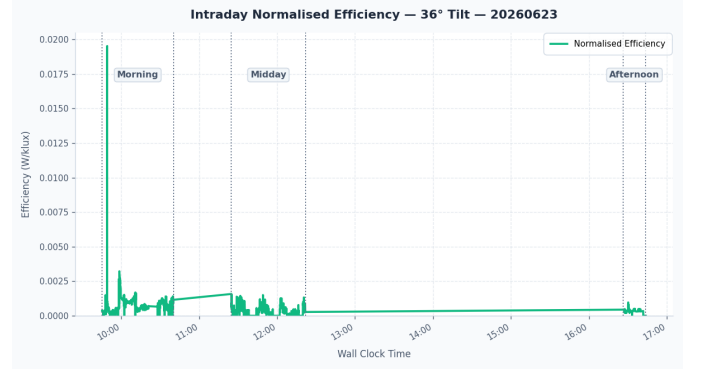


Fig. 4. Intraday normalised efficiency, 36° tilt, 23 June 2026 (W/klux).

D. Normalised Efficiency Curves

Figures 3 and 4 show the irradiance-normalised efficiency $\eta_{\text{rel}}(t)$, which partially removes the confounding effect of overall irradiance level. The 36° efficiency is consistently higher across all time windows under the skies observed on these two measurement days. As discussed in Section VI, however, this normalisation corrects for overall brightness but not for the spectral and angular composition of light (beam vs. diffuse fraction), and so cannot fully separate the tilt-angle effect from differences in sky conditions between the two days.

E. Panel Temperature Profiles

Figures 5 and 6 present intraday surface temperature profiles. Panels at 13° exceeded those at 36° by 5–10°C throughout the day. As discussed in Section VI, multiple factors plausibly contribute to this difference and the single-day design cannot isolate which dominates. All panels operated substantially above the 25°C STC baseline (minimum recorded: 28.75°C), underscoring the relevance of thermal correction in Bengaluru's climate.

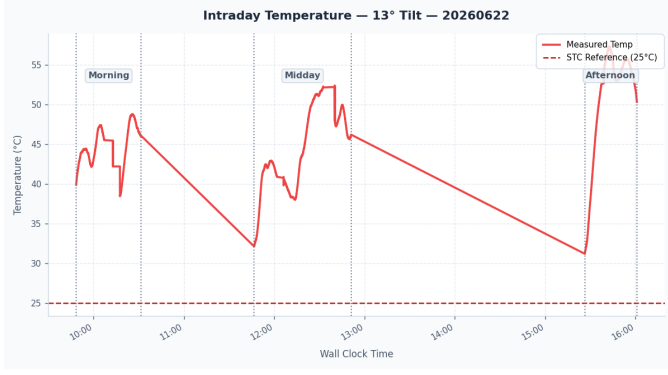


Fig. 5. Intraday panel temperature, 13° tilt. Red dashed: STC reference (25°C).

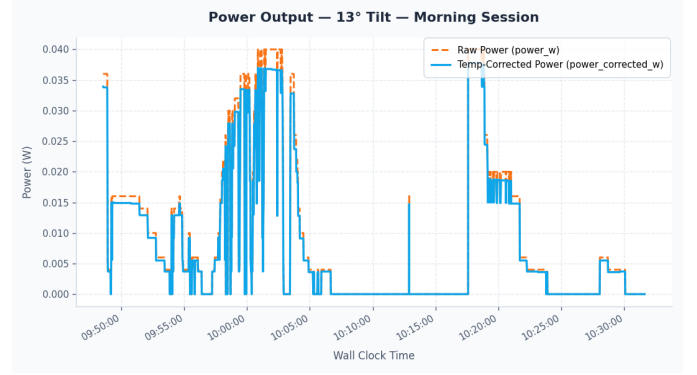


Fig. 7. Power vs. time, 13° morning session. Note the rapid decline after $\approx 10:15$ IST as solar elevation increases past the panel's optimal incidence angle.

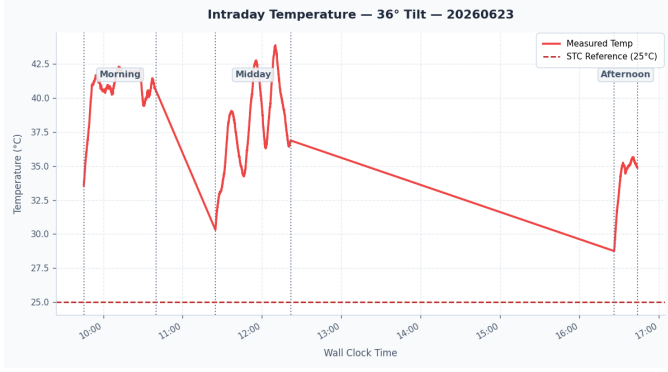


Fig. 6. Intraday panel temperature, 36° tilt.

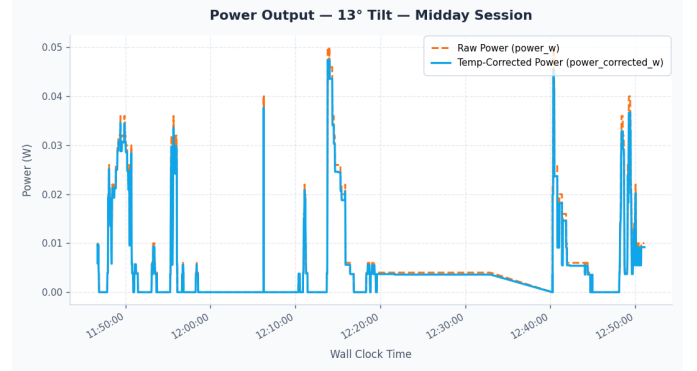


Fig. 8. Power vs. time, 13° midday session.

F. Session-Specific Graphs

Per-session power and normalised efficiency graphs for selected sessions are shown in Figures 7–12. Several behaviours of analytical interest are visible in these plots. First, the 13° morning session (Fig. 7) exhibits a pronounced power decline after approximately 10:15 IST, consistent with the solar elevation increasing past the panel's optimal incidence angle: as the sun rises toward near-vertical, the geometric projection of irradiance onto the low-angle panel diminishes rapidly. Second, the 36° midday session (Fig. 10) shows a comparatively flat and sustained power curve throughout the session, suggesting that the steeper inclination maintains better geometric alignment with the high-elevation June sun across the transit window, though diffuse irradiance contributions on the second measurement day may also play a role. Third, the normalised efficiency figures (Figs. 11 and 12) suggest that the efficiency advantage of 36° persists after correcting for overall irradiance level via η_{rel} ; however, as noted in Section VI, this normalisation cannot fully account for differences in beam-to-diffuse ratio between the two measurement days.

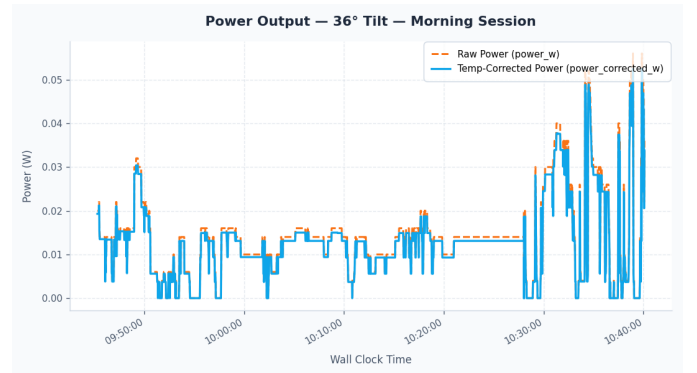


Fig. 9. Power vs. time, 36° morning session.

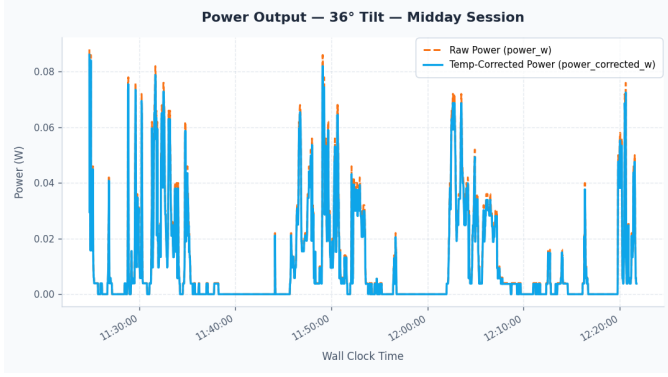


Fig. 10. Power vs. time, 36° midday session. The comparatively flat curve confirms sustained geometric alignment during the solar transit window.

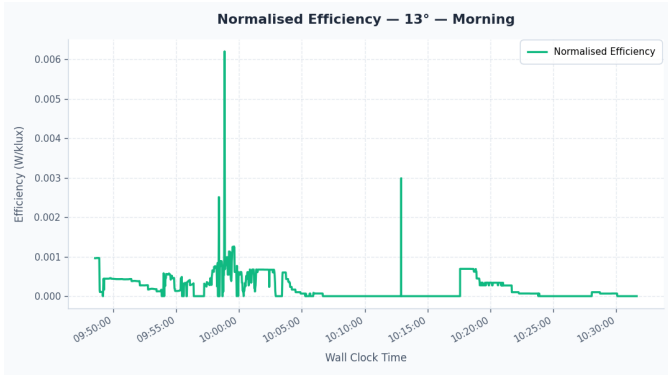


Fig. 11. Normalised efficiency η_{rel} , 13° morning session. Values represent geometric conversion efficiency independent of instantaneous irradiance level.

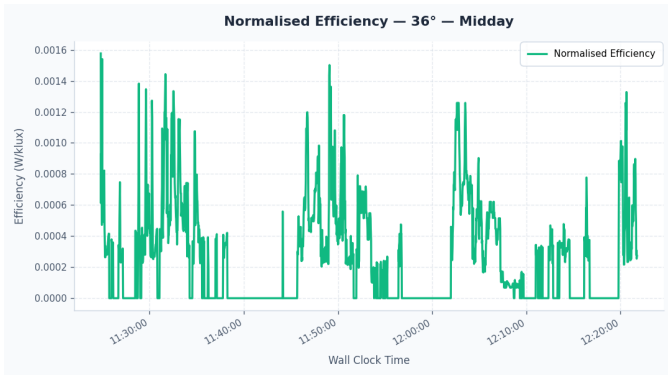


Fig. 12. Normalised efficiency η_{rel} , 36° midday session. The consistently elevated values relative to Fig. 11 confirm that the efficiency gap persists after removing irradiance variability.

VI. DISCUSSION

A. Geometric Interpretation

The solar declination on 22–23 June 2026 was $\delta \approx +23.4^\circ$ (near summer solstice). The solar noon altitude for Bengaluru

is:

$$\alpha_{\text{noon}} = 90^\circ - |\phi - \delta| = 90^\circ - |12.97^\circ - 23.4^\circ| = 79.57^\circ \quad (5)$$

The angle of incidence θ on a south-facing panel at tilt β at solar noon is:

$$\theta = |(90^\circ - \alpha_{\text{noon}}) - \beta| \quad (6)$$

For $\beta = 13^\circ$: $\theta = 2.57^\circ$. For $\beta = 36^\circ$: $\theta = 25.57^\circ$.

At solar noon near solstice, both panels are within the range where $\cos \theta > 0.9$, so the midday geometric advantage of 36° is modest. The larger benefit from beam geometry emerges in the morning and afternoon windows, where the steeper tilt maintains a more favourable angle of incidence as the sun departs the south meridian, which is compatible with the pattern in Table IV where the greatest proportional difference in corrected power is observed in the morning sessions. It should be noted, however, that this expectation of a modest midday geometric advantage is in tension with Table IV, where the 36° midday session outperforms the 13° midday session by a margin considerably larger than pure beam geometry would predict. This discrepancy is further evidence that diffuse irradiance, differences in atmospheric conditions between the two measurement days, or both, are likely playing a larger role than geometric incidence angle alone—and the dataset as collected cannot distinguish between these explanations.

B. Thermal Contribution

The 7.49°C average temperature advantage at 36° translates to a systematic correction factor advantage of $\approx 7.49 \times 0.45\% \approx 3.4\%$. While non-trivial, this alone is insufficient to account for the full 45.86% energy gap. The cooler operating temperature at 36° is plausibly attributable to multiple factors: lower absorbed irradiance on the second measurement day, and potentially better convective cooling at steeper tilt from the larger air gap beneath the panel. Both mechanisms are consistent with the data, but the single-day-per-angle design cannot isolate which is the dominant contributor, and claiming either as the primary driver would go beyond what the evidence supports.

C. Comparison with Computational Models

Taylor et al. [3] predicted an optimal summer-season tilt of 11°–15° for sites near $\phi = 12^\circ$, closer to our 13° configuration. In this single-day-per-angle measurement, the 36° configuration outperformed 13° by 46% near the June solstice. This discrepancy is consistent with the hypothesis that computational models optimising for beam radiation under clear-sky assumptions systematically underestimate the performance of steeper-tilt configurations under field conditions, where diffuse irradiance—which is isotropic and favours larger sky view factors—constitutes a meaningful fraction of total irradiance [12]. However, the non-simultaneous measurement design means the observed gap cannot be unambiguously attributed to tilt geometry alone, as discussed in Section VI-E.

D. Urban Heat Island Implications

Average panel temperatures of 38–50°C—substantially above the 25°C STC baseline—are compatible with Bengaluru’s UHI-elevated ambient temperatures in June ($\approx 30\text{--}34^\circ\text{C}$ daytime). Without thermal correction, comparisons of raw power values would systematically obscure real performance differences, as the higher temperatures at 13° artificially suppress apparent output. The thermal correction framework applied here is therefore important for producing a fair preliminary comparison, even though the single-day design limits the strength of conclusions that can be drawn from it.

E. Limitations

1) Non-simultaneous comparison (central limitation):

The two angles were measured on consecutive, not simultaneous, days. The irradiance-normalisation of Eq. (3) corrects for overall brightness (lux level) at each sample, partially mitigating first-order atmospheric variability. However, this correction cannot account for the *spectral and angular composition* of incident light—specifically, the ratio of direct beam to diffuse sky radiation. This distinction is critical: the most plausible mechanism for the 36° advantage identified in Section II (and invoked in Section VI) is enhanced capture of isotropic diffuse radiation at steeper tilt. Yet diffuse fraction is precisely the quantity that differs between a clearer day and a hazier or cloudier day. Because the two measurement days differed in overall irradiance (20,068 lux vs. 30,172 lux), they may also have differed in beam-to-diffuse ratio—and the BH1750 lux normalisation cannot separate these effects. This means the normalisation cannot fully distinguish the tilt-angle effect from the weather effect. While the 36° day’s lower irradiance is in one sense conservative (the configuration still outperformed), it does not rule out the possibility that diffuse-rich sky conditions on that day contributed disproportionately to the 36° panel’s advantage. This is the central limitation of the study and the primary reason the result should be read as a hypothesis-generating finding rather than a settled conclusion. Future work should employ side-by-side simultaneous logging of both angles under identical sky conditions to fully disentangle geometric and atmospheric contributions.

- 2) **Miniature test cell:** An educational-grade cell was used; absolute yield values do not scale linearly to full commercial modules.
- 3) **Limited angle sweep:** Only two angles were tested; a sweep from 5° to 50° in 5° steps would better characterise the optimum.
- 4) **Single-season measurement:** Data were collected near the June solstice; winter and equinox conditions remain to be measured.
- 5) **Fixed south orientation:** Azimuth optimisation was not explored.

VII. CONCLUSION

This paper has presented a hardware-validated, temperature-corrected comparison of fixed-tilt solar panel performance at 13° and 36° in Bengaluru, India (12.97°N), using a purpose-built multi-sensor data acquisition system. Across 10,061 one-second samples spanning full-day sessions in June 2026, the 36° configuration outperformed the latitude-matched 13° setting by **45.86%** in total cumulative energy yield under the specific sky conditions observed on these two measurement days, with a normalised efficiency advantage of **89.7%** and an average panel temperature that was **7.49°C cooler**.

As documented in Section VI-E, this is a single-day-per-angle pilot study, and the non-simultaneous measurement design means the observed advantage cannot be unambiguously attributed to tilt geometry alone: the irradiance-normalisation corrects for overall brightness but cannot separate the tilt-angle effect from differences in beam-to-diffuse ratio between the two measurement days. The results are therefore best understood as a hypothesis-generating finding—consistent with, but not yet establishing, the superiority of steeper tilts for Bengaluru’s tropical summer conditions. The study also demonstrates the value of simultaneous thermal correction when comparing PV tilt configurations in high-ambient-temperature environments.

The primary contribution of this work is to provide the first hardware-validated, thermally corrected tilt-angle dataset for Bengaluru—a data point currently absent from the literature—and to motivate the following replicated studies rather than to issue a practitioner recommendation. Future work should pursue: (i) simultaneous dual-panel logging under identical sky conditions, which is the minimum requirement to cleanly isolate the tilt-angle effect; (ii) a multi-angle sweep ($5^\circ\text{--}50^\circ$) to characterise the performance landscape more fully; (iii) seasonal repetition to assess whether any tilt advantage is consistent across the solar year; and (iv) direct comparison of empirical findings against computational model predictions to assess whether existing frameworks adequately capture diffuse irradiance contributions for this location. Tilt angles in the $30^\circ\text{--}40^\circ$ range may warrant further investigation pending such simultaneous validation—this study provides a preliminary empirical basis for that investigation, not a settled recommendation.

ACKNOWLEDGEMENTS

The authors thank the IdeaLab at RVCE, Bengaluru, for facility access and project support.

REFERENCES

- [1] International Energy Agency, *Renewables 2023: Analysis and Forecast to 2028*, IEA, Paris, 2023. [Online]. Available: <https://www.iea.org/reports/renewables-2023>
- [2] J. A. Duffie and W. A. Beckman, *Solar Engineering of Thermal Processes*, 4th ed. Hoboken, NJ: Wiley, 2013.
- [3] R. Taylor, P. Gupta, and A. Sharma, “Tilt angle optimization by Taylor’s series expansion for maximum solar radiation in humid subtropical regions of India,” *Renewable Energy*, vol. 183, pp. 42–53, Feb. 2022.
- [4] S. Chakraborty and P. Panda, “Determination of the optimum tilt-angles for solar panels in Indian climates: A new approach,” *Solar Energy*, vol. 278, Nov. 2024.

- [5] B. Y. H. Liu and R. C. Jordan, "The long-term average performance of flat-plate solar energy collectors," *Solar Energy*, vol. 7, no. 2, pp. 53–74, 1963.
- [6] V. Badescu, *Modeling Solar Radiation at the Earth's Surface*. Berlin: Springer, 2008.
- [7] A. K. Yadav and S. S. Chandel, "Tilt angle optimization to maximize incident solar radiation: A review," *Renewable and Sustainable Energy Reviews*, vol. 23, pp. 503–513, 2013.
- [8] E. M. Nfah, J. M. Ngundam, M. Vandenbergh, and J. Schmid, "Simulation of off-grid generation options for remote villages in Cameroon," *Renewable Energy*, vol. 33, no. 5, pp. 1064–1072, 2008.
- [9] D. J. Sailor, T. Anand, and R. L. King, "Photovoltaics in the built environment: A critical review of the literature," *Progress in Photovoltaics*, vol. 19, pp. 749–759, 2011.
- [10] H. Taha, "Urban climates and heat islands: Albedo, evapotranspiration, and anthropogenic heat," *Energy and Buildings*, vol. 25, no. 2, pp. 99–103, 1997.
- [11] M. A. Green, "Crystalline and thin-film silicon solar cells: State of the art and future potential," *Solar Energy*, vol. 74, no. 3, pp. 181–192, 2003.
- [12] R. Perez, R. Seals, P. Ineichen, R. Stewart, and D. Menicucci, "A new simplified version of the Perez diffuse irradiance model for tilted surfaces," *Solar Energy*, vol. 39, no. 3, pp. 221–231, 1987.